

ParcelVision: Monocular Camera-Based Real-World Object Dimensioning for Logistics Applications

Pranav D P | Akshay Bhat

[Project Repository & Source Code](#)

Abstract

ParcelVision is a research project aimed at building a software system capable of estimating the real-world dimensions and volumetric footprint of physical objects using only the camera of a consumer-grade smartphone. The intended application is the logistics and packers-and-movers industry, where accurate volume estimation directly determines truck allocation, labour planning, and final quotation pricing.

This report documents the full arc of the project: the problem that motivated it, the succession of technical approaches investigated, the fundamental theoretical barrier encountered, and the findings from each architectural approach evaluated. The central finding of this work is that dimension estimation from a camera is not a machine learning problem in the conventional sense. It is a scale recovery problem grounded in projective geometry, and solving it requires either external physical reference, multi-frame motion analysis, or hardware-level sensor data. No purely monocular, single-frame approach can resolve metric scale, and any system that claims otherwise is making an implicit assumption that the project has made explicit and rigorous.

1. Introduction

1.1 Background and Motivation

The logistics and household-moving industry in India and globally depends on human surveyors to assess shipment volume before generating a quotation. A surveyor visits the customer's premises, visually inspects furniture and belongings, estimates dimensions often without measuring tools, and produces a figure used to determine how many trucks, how many hours of labour, and ultimately what price to charge. This process is slow, inconsistent, prone to human error, and represents a significant operational bottleneck.

Quotation turnaround times of 24 to 48 hours are common. Mis-estimated volumes lead to either under-allocated trucks, causing multiple trips and increased cost, or over-allocated trucks, wasting capacity. Customer drop-off rates increase during the waiting period. The cost of errors falls on both the logistics company and the customer.

The natural question is whether a smartphone, present in nearly every customer's pocket, can replace or significantly assist the human surveyor. If a customer can point a phone at their sofa and receive an accurate volume estimate within seconds, quotation turnaround collapses from hours to minutes, errors reduce, and operational costs fall.

1.2 Project Objective

The objective of ParcelVision is to build a system that accepts video from a smartphone camera and produces, for each detected object in the scene, the following output:

Field	Description
object_id	A stable unique identifier for the detected object across frames
label	The semantic category of the object (e.g., sofa, refrigerator, cardboard box)
length	Estimated real-world length in metres
width	Estimated real-world width in metres
height	Estimated real-world height in metres
volume_m3	Computed volumetric footprint in cubic metres
confidence	A numeric confidence score for the measurement

The acceptable error margin for logistics purposes is plus or minus ten to twenty percent. This is a deliberately pragmatic target. The goal is not to match industrial LiDAR-grade measurement but to be accurate enough that quotations generated from these estimates are commercially viable. This distinction shaped every architectural decision in the project.

1.3 Scope of This Report

This report covers the complete intellectual history of the project. It documents not only what was built and what works, but also what was attempted and discarded, and the reasoning behind each decision. The succession of failed or rejected approaches is not a record of failure; it is the primary research output. Understanding why naive approaches do not work is what led the team to a technically sound architecture.

2. Problem Analysis

2.1 The Surface Problem and the Real Problem

On the surface, the problem appears to be an object detection problem: detect objects in images, then determine their size. This framing is misleading and was the source of early errors in the project.

Object detection models, including state-of-the-art neural networks, produce bounding boxes or segmentation masks in pixel coordinates. Pixel coordinates are two-dimensional projections of a three-dimensional world onto a flat image sensor. The projection process is lossy: it discards depth, and with depth, it discards absolute scale.

The real problem is not detection. The real problem is scale recovery: given a two-dimensional image or sequence of images, recover the absolute metric scale of the

scene. This is a problem in projective geometry, and it has a precise mathematical statement. A camera performs a projection from three-dimensional Euclidean space to a two-dimensional image plane. This projection is not invertible without additional information. Infinitely many three-dimensional scenes can produce the same two-dimensional image. A small object close to the camera and a large object far from the camera, in the correct proportions, produce identical pixel representations.

2.2 The Scale Ambiguity Theorem

This observation is not heuristic; it follows from the mathematics of the pinhole camera model. The image coordinates of a world point are given by:

Pinhole Camera Projection

$u = f_x \times (X/Z) + c_x$ $v = f_y \times (Y/Z) + c_y$ Where (X, Y, Z) is the world point, (f_x, f_y) is the focal length, (c_x, c_y) is the principal point, and (u, v) is the image coordinate. Critically, only the ratios X/Z and Y/Z appear. Any scaling of (X, Y, Z) by a common factor k produces the same (u, v) . This is the mathematical statement of scale ambiguity.

This means that a neural network trained on images cannot, by any amount of training data or architectural sophistication, recover metric scale from a single image unless it has learned an implicit prior about the size of objects from contextual cues. Such priors are unreliable, domain-specific, and do not generalise to novel objects, which is precisely what a logistics system must handle.

2.3 What Is Needed

Scale recovery requires at least one of the following:

- A known-size reference object visible in the image
- Known camera motion between two or more frames, with the motion expressed in metric units
- A depth sensor (structured light, time-of-flight, or stereo) that directly measures metric depth
- Strong prior knowledge about the class of object being measured, combined with statistical population data about its size

None of these are available for free in a generic smartphone camera system. The project's core research question is therefore: which of these constraints can be obtained automatically, cheaply, and without burdening the user?

3. System Components and Technology Decisions

3.1 Object Detection: YOLOv8-World

Object detection is the entry point of the pipeline. Given a video frame, the system must locate objects and classify them before any measurement can occur.

The project evaluated YOLOv8 and YOLOv8-World. Standard YOLOv8 is a closed-vocabulary detector trained on fixed categories. For logistics applications, this is insufficient: the system must handle sofas, dining tables, wardrobes, refrigerators, washing machines, cardboard boxes, and many other objects that may or may not appear in standard detection datasets.

YOLOv8-World extends the YOLO architecture with open-vocabulary detection capability, allowing the system to detect arbitrary object classes specified at inference time. This makes it suitable for logistics applications where the object vocabulary is broad and domain-specific. It provides bounding boxes, confidence scores, and class labels at real-time frame rates on modern mobile hardware.

The detection output feeds downstream components: segmentation, tracking, and depth estimation.

3.2 Segmentation: Segment Anything Model (SAM)

Bounding boxes from YOLO are rectangular regions of interest that inevitably include background pixels surrounding the object. For dimension estimation, background inclusion introduces noise into depth sampling and point cloud reconstruction. The system needs a precise mask of the object surface, not a rough bounding rectangle.

Meta's Segment Anything Model (SAM) is used to refine detection outputs into pixel-precise object masks. The pipeline feeds YOLO's bounding box as a prompt to SAM, which generates a segmentation mask isolating the object from its surroundings. This mask is used in all downstream processing.

SAM is large and computationally expensive. For production use, SAM2 or MobileSAM variants are preferable. The system includes a fallback: if the SAM mask is rejected due to low quality (measured by mask area relative to bounding box area, or by edge coherence), the bounding box is used directly.

3.3 Object Tracking

Video-based processing introduces the need for stable object identity across frames. Without tracking, the same physical object may receive different identifiers in consecutive frames, causing measurement instability and preventing temporal averaging.

The system implements IoU-based (Intersection over Union) tracking. Each detected object in a new frame is matched to tracked objects from previous frames by computing the overlap between their bounding boxes. Persistent IDs are assigned on first detection and maintained across frames. Temporal smoothing is applied to dimension estimates: the reported dimensions for each tracked object are a rolling average over recent frames, which suppresses per-frame noise.

3.4 Depth Estimation: DepthAnything V2

Depth estimation provides a per-pixel depth map from a single RGB frame. The project evaluated MiDaS and DepthAnything V2. DepthAnything V2 was selected for its superior performance on in-the-wild scenes and its robustness to lighting variation.

A critical and initially underestimated property of both models is that they produce relative depth, not metric depth. The output is a map of relative ordering: pixel A is closer than pixel B, and by approximately what factor. The absolute scale of these values has no physical meaning. A depth value of 0.5 from DepthAnything does not correspond to 0.5 metres or any fixed distance. This is the core limitation that drove the project's major architectural decisions.

3.5 Point Cloud Reconstruction

Given a depth map and known camera intrinsic parameters, the system lifts the 2D image into a 3D point cloud using the following transformation for each pixel (u, v) with depth value z:

Back-Projection Equations

$X = (u - cx) \times z / fx$ $Y = (v - cy) \times z / fy$ $Z = z$ Where fx , fy are the focal lengths and cx , cy is the principal point (optical centre) of the camera. These parameters are obtained from device calibration data or estimated from EXIF metadata.

The result is a set of 3D points representing the visible surface of the scene. An axis-aligned bounding box is fitted to the points belonging to each tracked object (as defined by the SAM mask), and the dimensions of this bounding box are the raw dimension estimates. Because the depth values are relative, these dimensions are in arbitrary units and must be scaled to produce metric measurements.

4. Approaches Investigated and Outcomes

The project systematically investigated multiple approaches to solving the scale recovery problem. Each is documented below with its rationale, implementation, and reason for rejection or partial acceptance.

4.1 Approach 1: Monocular Single-Frame Depth (Rejected)

The initial hypothesis was that combining YOLO detection with a depth estimation model would yield metric dimensions directly. The pipeline was implemented: YOLO produced bounding boxes, DepthAnything produced a depth map, the object region was sampled for depth values, and dimensions were computed from the spread of depth values.

The immediate observation was that dimensions varied dramatically as the camera was moved closer to or further from the object. The same sofa produced wildly different dimension estimates at different camera distances. The reason, as subsequently understood, is that DepthAnything produces relative depth. Moving closer increases the relative depth values without changing their absolute physical meaning.

This approach was rejected as a standalone solution. It was the discovery of its failure that prompted the deeper analysis of scale ambiguity documented in Section 2.

4.2 Approach 2: Reference Object Calibration (Rejected)

If scale ambiguity prevents metric estimation from an image alone, one solution is to place a known-size object in the scene and compute the scale factor from it. Common choices are a standard credit card (85.6 mm × 54 mm), an A4 sheet of paper (297 mm × 210 mm), or a coin.

The system would detect the reference object, compute the ratio of its known physical dimensions to its pixel span, derive a pixels-per-metre scale factor, and apply it to all depth-derived measurements.

This approach was implemented and demonstrated to produce correct measurements. However, it was rejected for production use in a logistics context on UX grounds. Requiring every user to carry a specific reference object and ensure it appears prominently in every frame is not a realistic expectation. The system would fail silently whenever the reference object was absent or partially occluded.

4.3 Approach 3: Manual Click-and-Measure Calibration (Temporary Workaround)

A simpler variant of reference calibration allows the user to manually identify two points in the frame and enter the real-world distance between them. The system computes the scale factor as the ratio of the entered distance to the pixel distance between the clicked points.

This approach works reliably and was implemented as a calibration tool during development. It serves as a ground-truth validator for testing other approaches. However, it is explicitly labelled a workaround. It requires user intervention, is not scalable, and produces poor user experience in a consumer-facing application. It was retained as a development utility and a fallback mode, not as the primary mechanism.

4.4 Approach 4: Structure from Motion (Evaluated, Fallback)

Structure from Motion (SfM) is a well-established technique in computer vision that recovers three-dimensional structure from a sequence of two-dimensional images by analysing the apparent motion of scene features as the camera moves.

The pipeline requires: extracting feature points from each frame (using ORB, SIFT, or similar), matching features across frames, estimating the relative camera motion between frames (rotation and translation up to scale), triangulating matched features to recover 3D positions, and bundle adjustment to refine the reconstruction.

SfM can recover structure up to a scale factor. To obtain metric scale, the translation between frames must be known in metric units. If the camera motion is known (from IMU sensors), metric scale can be recovered.

SfM was evaluated and found to produce good geometry with poor runtime performance. The compute requirements are high, latency is significant on mobile hardware, and the pipeline is brittle on textureless surfaces common in logistics scenes (white walls, plain furniture). SfM is retained as a potential offline processing approach and as the theoretical basis for the multi-frame motion pipeline.

4.5 Approach 5: WebXR and Visual Inertial Odometry (Evaluated)

Modern smartphones contain inertial measurement units (IMUs) consisting of accelerometers and gyroscopes. These sensors measure linear acceleration and angular velocity in the device frame. Integrating acceleration twice yields displacement, and integrating angular velocity yields orientation.

WebXR is a browser API that exposes AR capabilities including camera feed, IMU data, and device pose estimates. Apple's ARKit and Google's ARCore provide Visual Inertial Odometry (VIO), which fuses camera images and IMU readings to produce metric pose estimates of the camera in world space. Because VIO outputs are in metric units, they provide the scale reference needed to convert relative depth into metric depth.

This approach is theoretically the most elegant solution for a mobile-first deployment. The phone already knows how far it has moved in metric units; this information can be used directly to scale the depth estimates.

The evaluation revealed significant practical obstacles. WebXR support is inconsistent across browsers and device models. Many Android devices do not support ARCore or support it only partially. iOS support through WebXR is limited. The API surface is complex and documentation is sparse. Device-specific calibration of IMU sensor noise is required for accurate results. These factors prevented comprehensive validation across the range of target devices. However, commercial systems such as Apple's Measure app and Google's ARCore demonstrate that VIO-based metric measurement works in practice. WebXR/ARCore/ARKit therefore remains the most promising direction for future work, contingent on access to compatible hardware and browser environments.

4.6 Approach 6: Multi-Frame Motion with IMU Fusion (Attempted — Failed)

This approach was implemented and tested. Despite being theoretically well-grounded, it failed in practice. IMU drift on consumer smartphones was too severe to produce usable scale estimates. Sensor noise, integration errors, and the lack of reliable metric motion data made the approach non-viable as a standalone solution. This is now a rejected approach.

The core principle is that as the user moves the phone around an object to scan it, the camera undergoes motion. This motion creates parallax: features in the scene appear to shift in the image plane by amounts proportional to their distance. If the metric magnitude of the camera motion is known from IMU integration, and the pixel displacement of tracked features is known from optical flow, then the scale factor can be computed.

The scale recovery equation is:

Scale Recovery

$$\text{scale} = \frac{\text{real_motion_magnitude}}{\text{estimated_motion_magnitude_from_pixels}}$$
$$\text{metric_depth} = \text{relative_depth_from_model} \times \text{scale}$$
Where $\text{real_motion_magnitude}$ is derived from IMU integration (metres) and $\text{estimated_motion_magnitude_from_pixels}$ is derived from feature tracking in image coordinates combined with depth model outputs.

This approach fuses two streams of information: the geometric stream from the camera (depth model, feature tracking, optical flow) and the physical stream from the IMU (acceleration, gyroscope readings). Neither stream alone provides metric scale; together they constrain it.

4.7 Approach 7: Metric Depth Models (ZoeDepth / Metric3D) (Evaluated, Rejected)

Following the failure of IMU fusion, the project evaluated metric depth models—specifically ZoeDepth and Metric3D—as a potential alternative path to scale recovery. These models are trained to predict metric depth directly from a single image, rather than the affine-invariant relative depth produced by DepthAnything V2. The appeal was significant: if a model could reliably output depth in metres without any external reference, the scale recovery problem would dissolve.

Experiments with ZoeDepth and Metric3D revealed that real-world dimension estimates were not sufficiently reliable for logistics applications. Accuracy varied substantially across scene types and was heavily dependent on the training distribution of each model. Scenes closely resembling the training data produced plausible metric estimates, while novel household scenes—the primary use case for ParcelVision—produced estimates with errors far exceeding the ten to twenty percent target margin. Distribution shift remained a fundamental and persistent concern: the models could not generalise consistently to the variety of environments and object configurations encountered in real logistics contexts.

The theoretical analysis in Section 7 explains this outcome: metric depth models encode implicit scene priors during training, and these priors are inherently domain-specific. Unlike relative depth models, which make no claim about absolute scale and are therefore stable and generalisable, metric depth models make an absolute claim that can be wrong in ways that are difficult to detect at inference time. For a system required to provide reliable measurements without user verification, this silent failure mode is not acceptable.

Metric depth models were therefore not adopted as the final solution. Future foundation models trained on substantially larger and more diverse datasets may narrow the distribution shift problem, making this class of approach worth revisiting. However, based on evaluation at the time of this project, the reliability ceiling was insufficient for logistics-grade dimension estimation.

5. Current Architecture

5.1 Pipeline Overview

The implemented pipeline processes video frames in sequence and maintains state across frames for tracking and temporal averaging. The pipeline consists of the following stages: YOLO-World detection, SAM segmentation, object tracking, relative depth estimation, point cloud reconstruction, and 3D bounding box fitting. Automatic metric scale recovery is not yet resolved; the system currently produces relative

dimensions requiring calibration for metric output. The following table describes each stage:

Stage	Description
Video Input	A video stream captured by the smartphone camera at a stable frame rate (target 30 fps). The user moves the phone slowly around the object being measured.
YOLO-World Detection	Each frame is processed by YOLOv8-World to detect and classify objects. Bounding boxes and confidence scores are produced for each detected object in the frame.
SAM Segmentation	Detected bounding boxes are used as prompts to Segment Anything Model, which produces pixel-precise object masks. Low-quality masks fall back to bounding box regions.
Object Tracking	Detected objects are matched to the persistent tracker using IoU matching. New objects are assigned IDs; existing objects maintain their IDs. Dimension estimates are smoothed temporally.
DepthAnything V2	A relative depth map is computed for the full frame. Object regions (defined by SAM masks) are sampled from this depth map to obtain relative depth values for each object.
Point Cloud Reconstruction	The back-projection equations are applied to the relative depth map to generate a 3D point cloud for each object. Dimensions produced are in relative units; metric conversion requires a scale factor supplied via manual calibration or a validated automatic method.
3D Bounding Box	An axis-aligned bounding box is fitted to each object's metric point cloud. The box dimensions (length, width, height) are the estimated real-world measurements.
Volume and Output	Volume is computed as length times width times height. The full output record is assembled and returned for each tracked object.

5.2 Implementation Status

Component	Status	Notes
YOLO-World Detection	Fully implemented	Running at inference speed on test hardware. Open-vocabulary detection operational.
Object Tracking	Fully implemented	IoU tracking with persistent IDs and temporal smoothing operational.
SAM Segmentation	Fully implemented	Integrated with YOLO; fallback to bounding box on weak masks.
DepthAnything V2 Depth	Fully implemented	Relative depth maps generated per frame. Relative-only limitation confirmed.

Point Cloud Reconstruction	Partially implemented	Back-projection operational; metric accuracy depends on scale factor.
3D Bounding Box Fitting	Partially implemented	Works correctly given accurate point cloud; sensitivity to noise noted.
Manual Calibration	Implemented (dev tool)	Click-to-calibrate functional; used as ground truth validator.
Optical Flow Tracking	Explored	ORB and Lucas-Kanade explored and tested as part of IMU fusion experimentation.
IMU Fusion	Implemented and Evaluated / Rejected	Sensor access, Kalman filter design, and integration implemented and evaluated. Rejected due to excessive drift on consumer smartphone hardware.
Scale Recovery (auto)	Unresolved	The project investigated all practically available monocular scale recovery approaches within project constraints. None produced a sufficiently accurate, device-agnostic, and user-friendly solution. Remains unresolved for fully automatic operation.
WebXR Integration	Explored	Prototyped; device support constraints identified. Targeted at supported devices only.
Structure from Motion	Explored	Evaluated as offline fallback; runtime too high for real-time use.
Metric Depth Models (ZoeDepth / Metric3D)	Explored and Evaluated / Rejected	Evaluated as a potential scale-free alternative. Insufficient reliability due to distribution shift; rejected for logistics use.

5.3 The Critical Path

The single blocking issue for the entire system is automatic metric scale recovery. All other components are functional. The system can detect, segment, track, and produce relative depth. It cannot yet convert relative depth to metric depth without user intervention.

The project investigated all practically available monocular scale recovery approaches accessible within project constraints. None produced a sufficiently accurate, device-agnostic, and user-friendly solution. IMU fusion was implemented, evaluated, and rejected due to drift on consumer smartphone hardware. Metric depth models were explored and found insufficient for logistics-grade accuracy. The most promising remaining direction is WebXR-based Visual Inertial Odometry on devices with full

ARCore/ARKit support, which could not be fully validated due to hardware constraints during the project.

6. Evaluation Methodology

6.1 Ground Truth Collection

Accurate evaluation requires ground truth dimension measurements. Ground truth is collected using a physical tape measure for each test object. A set of test objects spanning the likely logistics vocabulary was assembled: a two-seater sofa, a dining chair, a rectangular dining table, a standard cardboard box, and a wardrobe. Dimensions were measured to the nearest centimetre.

6.2 Metrics

The primary evaluation metric is Mean Absolute Percentage Error (MAPE) per dimension:

MAPE Definition

$$\text{MAPE} = (1/n) \times \sum | \text{estimated} - \text{actual} | / \text{actual} \times 100\%$$
 Computed separately for length, width, and height. Volume error is also computed using the same formula applied to the volume estimate versus the ground truth volume (product of measured dimensions).

Secondary metrics include temporal stability (variance of dimension estimates across frames for a static object) and detection reliability (fraction of frames in which the object is correctly detected and segmented).

6.3 Calibration Baseline

With manual click-to-calibrate scale correction, the system achieves dimension estimates within approximately eight to fifteen percent of ground truth on the test object set, depending on object geometry and surface texture. This establishes the calibration baseline and confirms that the remainder of the pipeline (detection, segmentation, depth sampling, point cloud, bounding box fitting) is sound. The remaining challenge is entirely the automatic scale recovery problem.

7. Theoretical Analysis

7.1 Why Monocular Depth Cannot Provide Metric Scale

The theoretical impossibility of recovering metric scale from a single monocular image without a reference is worth making precise, because it has practical implications for the entire field of single-image 3D reconstruction.

Consider a neural network trained to predict depth from RGB images. During training, it observes pairs of (image, depth map) where the depth maps are obtained from

LiDAR or stereo sensors and are in metric units. The network learns statistical associations between image features (texture gradients, relative object sizes, perspective cues, defocus blur) and depth values. These associations encode implicit priors about the world: chairs are typically about 80 cm tall, sofas are about 90 cm tall, walls have predictable texture gradients.

When such a network is applied to an image outside its training distribution, these priors may not hold. A miniature sofa would be assigned a depth map consistent with a full-size sofa at a corresponding distance. The network has no mechanism to distinguish these cases from image data alone.

DepthAnything V2 avoids this trap by explicitly not predicting metric depth. It predicts relative depth (affine-invariant depth), which is stable across scales and distances. This makes it a more reliable model, but one that definitively cannot produce metric measurements. It is the right tool for what it does; it simply does not solve the scale recovery problem.

7.2 Why IMU Fusion Works in Principle

The IMU fusion approach works because it provides exactly the piece of information that the camera geometry cannot. The camera tells us about the angles and relative distances in a scene. The IMU tells us about the physical motion of the device in metric units. Together, they over-constrain the scale.

Mathematically, if the camera moves by a known displacement d in metric space, and this produces an observed pixel displacement p in the image, then the depth z of the feature being tracked satisfies:

Scale Constraint from Motion

$z = f \times d / p$ Where f is the focal length in pixels. This is the classic triangulation equation. The metric displacement d from the IMU and the pixel displacement p from optical flow together determine z in metric units, from which the full scale factor for the relative depth map is derived.

The practical difficulty is that IMU integration accumulates drift, and the scale estimate has variance that must be managed through filtering. These are engineering challenges with known solutions, not theoretical obstacles.

In practice, however, these engineering challenges proved intractable on consumer hardware during testing. IMU integration drift exceeded acceptable thresholds for the scan durations required, and no reliable filtering strategy was found to compensate. The approach was abandoned.

7.3 Error Sources and Propagation

The final dimension estimate is a product of multiple noisy estimates. The error sources and their approximate contributions are:

Error Source	Nature and Magnitude
Detection bounding box accuracy	Depends on YOLO confidence; typically 1-5% error on well-lit objects.
SAM segmentation quality	Depends on surface texture and lighting; errors in mask boundaries propagate to point cloud.
Relative depth model accuracy	DepthAnything V2 produces accurate relative ordering with some non-linearity; approximately 5-10% shape error.
Scale factor estimation	The dominant error source currently. IMU integration drift and optical flow noise contribute jointly.
Point cloud bounding box fitting	Sensitive to outlier depth points; robust fitting methods (RANSAC on bounding box) partially mitigate this.
Camera intrinsic parameters	Errors in focal length or principal point propagate linearly to dimension estimates.

Total system error is a convolution of these individual errors. The calibration baseline result (eight to fifteen percent MAPE with manual calibration) indicates that removing the scale recovery error source alone is sufficient to bring the system within the target accuracy band.

8. Related Work and Comparison

8.1 Commercial Systems

Apple's Measure app uses ARKit, which implements Visual Inertial Odometry, to provide metric distance measurements on compatible iOS devices. It requires the user to explicitly mark start and end points of a measurement and is designed for single linear distances rather than volumetric object measurement. It does not integrate object detection or automatic object boundary identification.

Google's ARCore provides similar capabilities on Android. Neither platform exposes the underlying VIO pose data in a form readily accessible to web applications, which is the source of the WebXR accessibility constraints encountered in this project.

Industrial solutions such as Matterport use structured light sensors, not cameras alone, and are not relevant comparators for a smartphone-camera-only system.

8.2 Academic Literature

Significant academic work exists on monocular metric depth estimation. Models such as ZoeDepth and Metric3D attempt to predict metric depth directly by leveraging large-scale training data with metric labels and explicit scene-scale conditioning. These represent the state of the art in single-image metric estimation and were evaluated in this project as part of the systematic investigation of scale recovery approaches (see Section 4.7).

The practical limitation of metric depth models is their dependence on training distribution. They perform well on scenes similar to their training data and poorly on

novel scenes. For a logistics system that must handle any combination of furniture in any home environment, distribution shift is a genuine risk.

Visual Simultaneous Localisation and Mapping (vSLAM) systems such as ORB-SLAM3 and OpenVINS represent the academic baseline for multi-frame metric reconstruction. These systems solve exactly the problem this project requires but are designed for robotics applications with GPU compute budgets and low-latency requirements different from mobile consumer applications.

9. Future Work

9.1 Immediate Next Steps

- Evaluate WebXR-based Visual Inertial Odometry on devices with full ARCore or ARKit browser support, validating metric accuracy against the ground truth test object set.
- Evaluate ARCore and ARKit native implementations (outside the browser context) to determine whether native VIO access yields sufficient metric accuracy for logistics measurement.
- Evaluate LiDAR-assisted measurement on supported hardware (iPhone Pro, iPad Pro) as a hardware-guaranteed scale recovery path for premium device deployments.

9.2 Medium-Term Development

- Investigate future metric depth foundation models trained on substantially larger and more diverse datasets, reassessing whether distribution shift can be reduced to acceptable levels for logistics measurement.
- Develop a user interface that guides the user through the scanning motion, providing feedback when sufficient parallax has been accumulated for reliable scale estimation.
- Extend the object vocabulary with logistics-specific fine-tuning of the detection model on a curated dataset of common household items.
- Implement automatic quote generation from aggregated volume measurements, integrating with a logistics pricing engine.

9.3 Long-Term Research Directions

- Investigation of few-shot size priors: using database knowledge of common object dimensions as a soft constraint on the scale estimate, improving robustness in low-motion scenarios.
- Explore hybrid systems combining learned depth priors with geometric constraints from multi-frame motion, potentially providing robustness where either stream alone fails.
- Multi-device or stereo capture: using a second camera to provide a stereo baseline, recovering scale from disparity without requiring either IMU fusion or hardware depth sensors.

10. Conclusion

ParcelVision set out to answer a commercially motivated question: can a consumer smartphone estimate real-world object dimensions accurately enough to replace human surveyors in logistics quotation workflows? The answer, based on this project's research, is yes in principle, but with significant engineering requirements that are not obvious from a naive problem framing.

The central contribution of this project is the precise identification and characterisation of the scale recovery problem as the fundamental barrier in monocular dimension estimation. Single-image, single-model approaches are theoretically incapable of recovering metric scale. Any system that appears to do so is either using an implicit reference (scene context, known object priors) or exploiting multi-frame information.

The project implemented and evaluated all components of the measurement pipeline. The calibration baseline demonstrates that the pipeline from detection through to point cloud and bounding box fitting is sound and capable of meeting the target accuracy when scale is provided. Multiple scale recovery approaches were evaluated: IMU fusion with optical flow was implemented and found non-viable on consumer hardware due to sensor drift; metric depth models were evaluated and found insufficient due to distribution shift and unreliable generalisation. WebXR-based Visual Inertial Odometry remains the most promising direction but could not be fully validated due to hardware constraints during the project. LiDAR-assisted measurement on supported devices represents a viable hardware path for premium deployments.

The architecture arrived at through this process is not the naive architecture one would propose at the outset. It integrates geometry, motion, and physical sensor data in a way that respects the mathematical constraints of the problem. ParcelVision is not an AI system that guesses sizes. It is a measurement system that uses computer vision, projective geometry, inertial sensing, and statistical filtering to recover physical measurements from a device that was not designed for measurement.

The primary contribution of this project is the systematic investigation of scale recovery methods for monocular smartphone-based dimension estimation, and the precise identification of their theoretical and practical limitations. Several approaches succeeded under constrained conditions; several others failed due to engineering or fundamental obstacles. No approach produced a fully automatic, reliable, and device-agnostic metric measurement system using only a standard monocular camera. This outcome is itself a research finding: it establishes the boundary of what is achievable without additional hardware or privileged sensor access, and points clearly toward WebXR/ARCore/ARKit as the most promising near-term path. ParcelVision is not an AI system that guesses sizes. It is a systematic research investigation that determined how and why metric measurement from monocular vision is hard, which approaches are viable, and where future work should focus.

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